

2019 Mathematics Kangaroo Contest

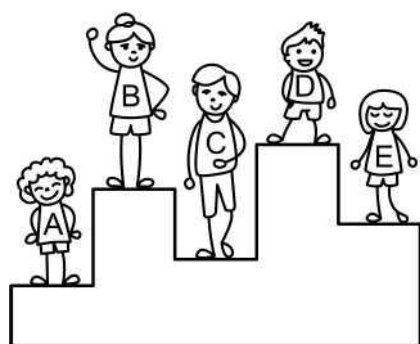
—Ecolier—

2019 数学袋鼠竞赛

Part 1: 8 problems, 3 points each | 第一部分: 8 道题目, 每题 3 分

- 1 • The higher the step on the podium, the higher the rank of the runner. Who finished third?

领奖台上站的台阶越高, 跑步者的排名越靠前。问谁是第三个跑完的?



(A) A

(B) B

(C) C

(D) D

(E) E

- 2 • In the pictures, each dot stands for 1 and each bar stands for 5. For example,  stands for

8. Which picture stands for 12?

在图中, 每个点代表 1, 每个长条代表 5。例如  代表 8。问哪张图片代表 12?



3. Yesterday was Sunday. What day is tomorrow?

昨天是星期天。明天是星期几?

(A) Tuesday 星期二

(B) Thursday 星期四

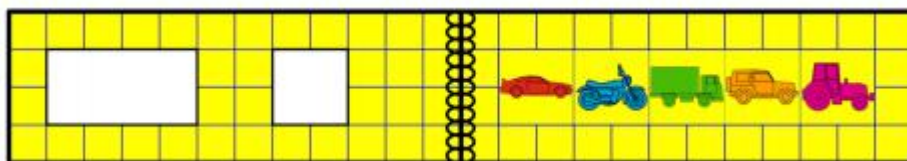
(C) Wednesday 星期三

(D) Monday 星期一

(E) Saturday 星期六

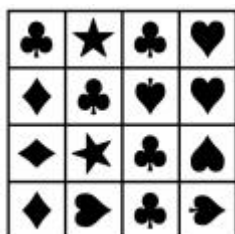
- 4 • There are two holes in the cover of a book. When the book is open, it looks like this: Which pictures does Olaf see through the holes when he closes the book?

书的封面上有两个洞。当这本书打开时，它看起来像这样：当 Olaf 把这本书合上时，他会看到哪些图片？



- (A) , , (B) , , (C) , ,
 (D) , , (E) , ,

5.



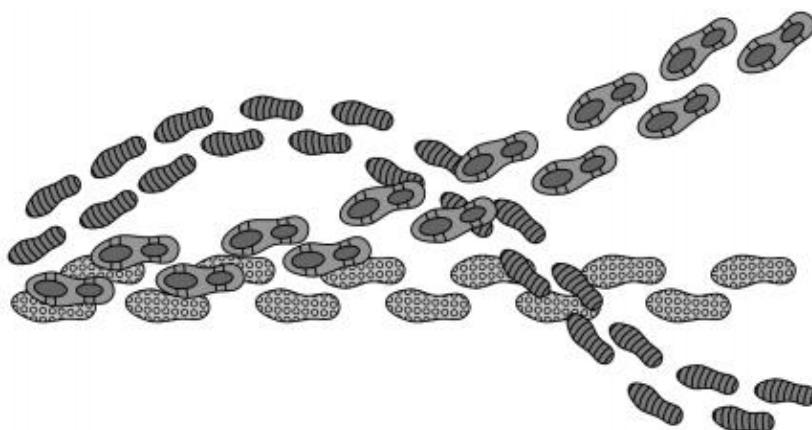
Karina cuts out one piece like this from the sheet above. Which piece can she get ?

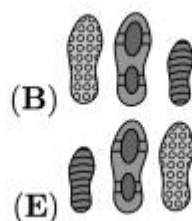
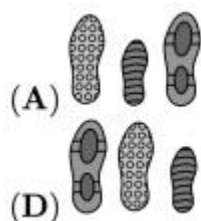
Karina 从上图中切出像 这样的一片，那么哪一块儿是她有可能得到的？

- (A) (B) (C)
 (D) (E)

6 • Three people walked across a field of snow wearing muddy shoes. In which order did they do this?

三个人穿着泥泞的鞋子走过一片雪地。他们是按什么顺序走的？



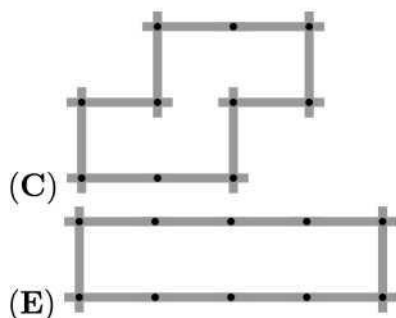
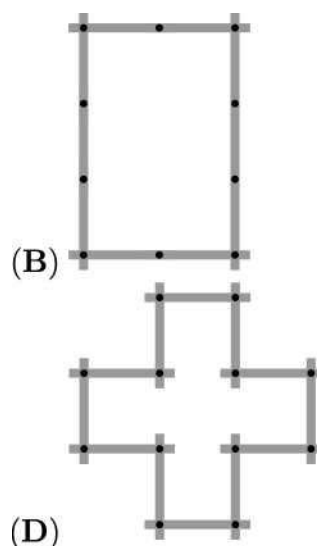
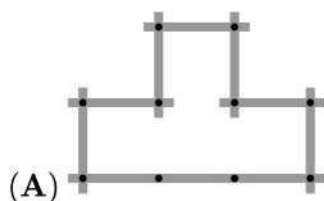


7.



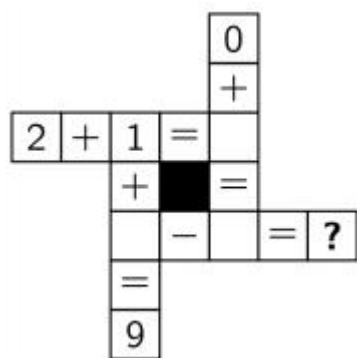
Pia makes shapes with the connected sticks shown above in the picture. Which of the following shapes needs more sticks than Pia has?

Pia 将木棒连接在一起形成如上图所示的形状。组成下面哪个形状需要使用比 Pia 现在有的还多的木棒？



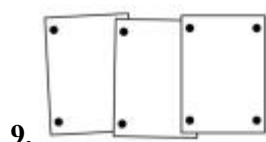
8 • What number should replace the question mark when all the calculations are completed correctly?

如果所有的计算都正确的话，哪个数应该代替问号？



- (A)4 (B)5 (C)6 (D)7 (E)8

Part 2: 8 problems, 4 points each | 第二部分: 8 道题目, 每题 4 分

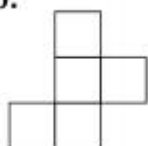


Linda pinned up 3 photos in a row on a cork board using 8 pins. Peter wants to pin up 7 photos in the same way. How many pins does he need?

Linda 在软木板上使用 8 个大头针在一行上固定了 3 张照片。Peter 想以同样的方式固定 7 张照片。他需要多少个大头针?

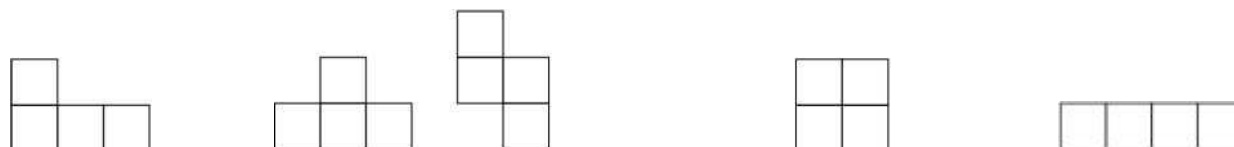
- (A) 14 (B) 16 (C) 18 (D) 22 (E) 26

10.



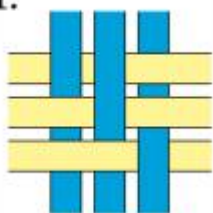
Dennis wants to remove one cell from the above shape. How many of the following shapes can he get?

Dennis 想从上面的形状中删除一个单元格。他可以获得下面形状中的几种?



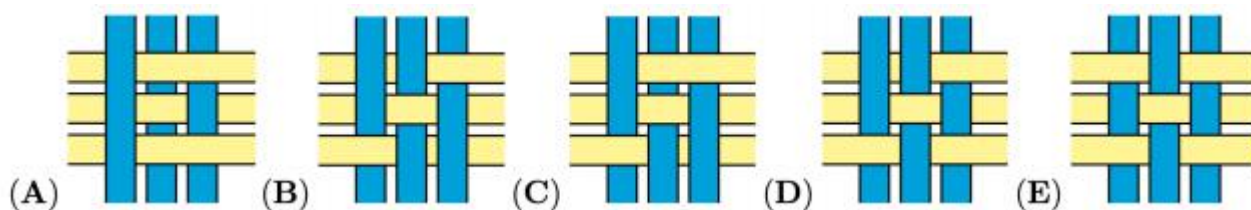
- (A)1 (B)2 (C)3 (D)4 (E)5

11.



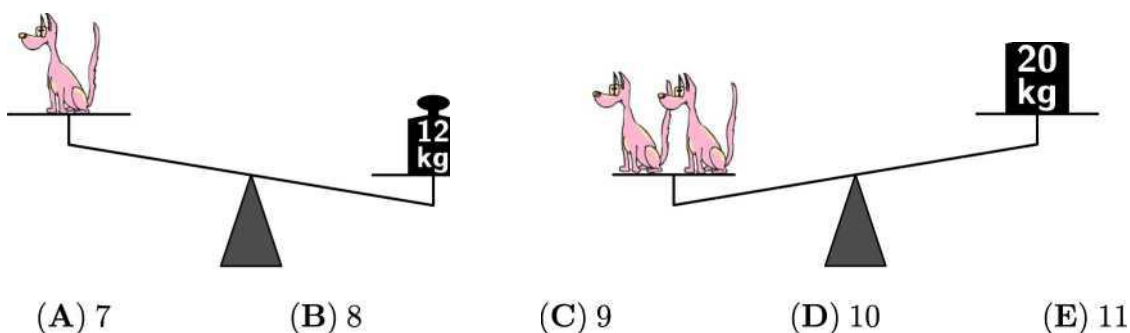
Six strips are woven into a pattern as shown above. What does the pattern look like from the back?

如上图所示，六条带子编织成一个图案。这个图案从背面看是什么样？



12 • The weight of dog toy is a whole number in kilograms. How many kilograms does one dog toy weigh? ("kg" is the abbreviation for kilograms.)

玩具狗的重量是整数千克。问一只玩具狗的重量是多少千克？(kg 是千克的简称。)



13 • Sara has 16 blue marbles. She can trade marbles in two ways: 3 blue marbles for 1 red marble or 2 red marbles for 5 green marbles. What is the maximum number of green marbles she can get?

Sara 有 16 个蓝色玻璃球。她可以通过两种方式交换玻璃球：3 个蓝色玻璃球可以换 1 个红色玻璃球，或者 2 个红色玻璃球可以换 5 个绿色玻璃球。问她最多可以获得多少个绿色玻璃球？

(A) 5 (B) 10 (C) 13 (D) 15 (E) 20

14.

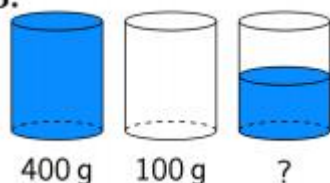
$$\square \square \square + \square ?$$

Steven wants to write each of the digits 2, 0, 1 and 9 in one of the boxes of the sum. He wants to get the largest possible answer. Which digit could he write instead of the question mark?

Steven 想要将数字 2, 0, 1 和 9 分别填在加法算式的每一个方框中。他希望获得尽可能大的答案。那么他可以将问号替换为哪个数字?

- (A) Either 0 or 1 0 或者 1 (B) Either 0 or 2 0 或者 2 (C) Only 0 只能是 0
 (D) Only 1 只能是 1 (E) Only 2 只能是 2

15.



A full glass of water weighs 400 grams. An empty glass weighs 100 grams. How many grams does a half-full glass of water weigh?

装满水的杯子重 400 克。一个空杯重 100 克。那么装有半杯水的杯子重多少克?

- (A) 150 (B) 200 (C) 225 (D) 250 (E) 300

16.



Together we
cost 5 cents.
我们一起
值 5 美分

- (A) 8 cents 美分
(D) 11 cents 美分



Together we
cost 7 cents.
我们一起
值 7 美分

- (B) 9 cents 美分
(E) 12 cents 美分



Together we
cost 10 cents.
我们一起
值 10 美分



How much do
cost together?
我们在一起
值多少钱?

- (C) 10 cents 美分

Part 3: 8 problems, 3 points each 第三部分: 8 道题目 9 每题 5 分
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17 • Each shape stands for a different number. The sum of the three numbers in each row is shown to the right of the row. Which number does the ★ stand for ?

每个形状代表不同的数。每行中三个数的总和显示在行的右侧。那么 ★ 代表几？

			15
			12
			16

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

18 • Anna used 32 small white squares to frame a 7 by 7 picture. How many of these small white squares does she need to frame a 10 by 10 picture?

Anna 使用了 32 个白色小方块来给一张 7 乘 7 的画加框。那么给 10 乘 10 的画加框，她需要多少个白色小方块？



- (A) 36 (B) 40 (C) 44 (D) 48 (E) 52

19 • The pages of a book are numbered 1, 2, 3, 4, 5 and so on. The digit 5 appears exactly 16 times. What is the maximum number of pages this book could have?

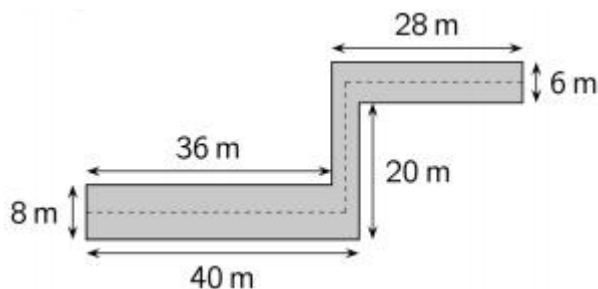
书的页面的编号为 1, 2, 3, 4, 5, 并一直继续。数字 5 恰好出现了 16 次。这本书最多可以有多少页？

- (A) 49 (B) 64 (C) 66 (D) 74 (E) 80

20. A hallway has the dimensions shown in the picture. A cat walks on the dashed line along the

middle of the hallway. How many meters does the cat walk? (m stands for meter.)

走廊的尺寸如图所示。一只猫在走廊中间的虚线上走。问猫走了多少米? (m 表示米。)



(A) 63

(B) 68

(C) 69

(D) 71

(E) 83

21. In a park there are 15 animals: cows, cats and kangaroos. We know that precisely 10 are not cows and precisely 8 are not cats. How many kangaroos are in the park?

在公园里有 15 只动物：牛，猫和袋鼠。我们很准确地知道，恰好有 10 只不是牛，恰好有 8 只不是猫。问公园里有多少只袋鼠？

(A) 2

(B) 3

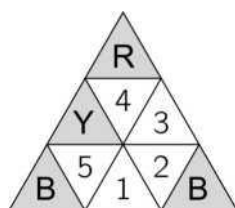
(C) 4

(D) 8

(E) 10

22 • Mary has 9 small triangles: 3 of them are red (R), 3 are yellow (Y) and 3 are blue (B). She wants to form a big triangle by putting together these 9 small triangles so that any two triangles with an edge in common are different colours. Mary places some small triangles as shown in the picture. Which of the following statements is true after she has finished?

Mary 有 9 个小三角形：其中 3 个是红色(R)，3 个是黄色(Y)，3 个是蓝色(B)。她希望将这 9 个小三角形组合在一起形成一个大三角形，任何两个有公共边的小三角形都是不同的颜色。如图所示，Mary 放置了一些小三角形。在完成后，以下哪项陈述是正确的？



(A) 1 is yellow and 3 is red

1 是黄色而 3 是红色

(B) 1 is blue and 2 is red

1 是蓝色而 2 是红色

(C) 1 and 3 are red

1 和 3 是红色

(D) 5 is red and 2 is yellow

5 是红色而 2 是黄色

(E) 1 and 3 are yellow

1 和 3 是黄色

23 • One of five children Alek, Bartek, Czarek, Darek and Edek has eaten a cookie.

Alek says: "I haven't eaten a cookie."

Bartek says: "I have eaten a cookie."

Czarek says: "Edek hasn't eaten a cookie."

Darek says: "I haven't eaten a cookie."

Edek says: "Alek has eaten a cookie."

Only one child lies. Who has eaten the cookie?

在 Alek, Bartek, Czarek, Darek 和 Edek 五孩子中, 有一人吃了一个饼干。

Alek 说: "我没有吃过饼干。"

Bartek 说: "我吃了饼干。"

Czarek 说: "Edek 没有吃过饼干。"

Darek 说: "我没有吃过饼干。"

Edek 说: "Alek 吃了饼干。"

只有一个孩子在撒谎。问谁吃了饼干?

(A) Alek

(B) Bartek

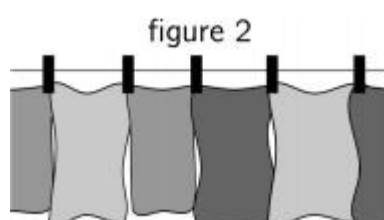
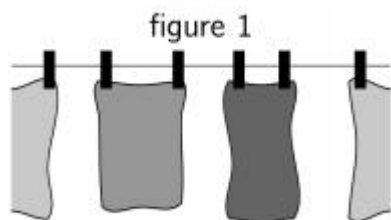
(C) Czarek

(D) Darek

(E) Edek

24 • Emil started to hang up towels using two pegs for each towel as shown in figure 1. He realised that he would have not enough pegs and began to hang up the towels as shown in figure 2. Overall, he hung up 35 towels and used 58 pegs. How many towels did Emil hang up in the way shown in figure 1?

如图 1 所示, 开始时 Emil 每条毛巾用两个夹子挂起。当他发现他没有足够的夹子时, 他开始用如图 2 所示的方式挂毛巾。总的来说, 他挂了 35 条毛巾, 并使用了 58 个夹子。问按照如图 1 所示的方式 Emil 挂了多少条毛巾?



(A) 12

(B) 13

(C) 21

(D) 22

(E) 23

答案: ECADA ADBBC CEBAD DECBE BEBD